Math 54 — Week 6

Linearly independent sets and bases; dimension and rank (Lay 4.3, 4.5)

§1 Facts used from Weeks 1–5

(W1) (Week 3) A set of two or more vectors is dependent exactly when one of them is a combination of the others. More than n vectors in $\mathbb{R}^n$ are dependent.

(W2) (Weeks 2–3) The columns of A are independent $\iff$ a pivot in every column; they span $\mathbb{R}^m$ $\iff$ a pivot in every row. A homogeneous system with more unknowns than equations has a nontrivial solution.

(W3) (Week 1) Row operations do not change the solution set of $A\mathbf{x} = \mathbf{0}$, and each can be undone by a row operation.

(W4) (Week 5) Subspaces; a span is a subspace; $\operatorname{Nul} A$, $\operatorname{Col} A$, $\operatorname{Row} A = \operatorname{Col} A^T$. The parametric vector form of $A\mathbf{x} = \mathbf{0}$ gives a linearly independent spanning set for $\operatorname{Nul} A$, one vector per free variable.

(W5) (Weeks 4–5) Invertible matrix theorem, including $\det A \neq 0$. Each pivot lies in a different row and column, so an $m \times n$ matrix has at most $\min(m, n)$ pivots.

§2 This week

§2.1 Linearly independent sets; bases (Lay 4.3)

The question. The five columns of the matrix below span $\operatorname{Col} A$, but some of them are redundant. Which can be dropped, what is the smallest kind of set that still spans, and how do we find one for $\operatorname{Nul} A$, $\operatorname{Col} A$ and $\operatorname{Row} A$?

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

Definition. A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in a vector space V is **linearly independent** if $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ only for $c_1 = \cdots = c_p = 0$, and **linearly dependent** otherwise.

Theorem (Characterization of dependent sets (Lay 4.3, Thm. 4))
An indexed set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some $\mathbf{v}_j$ with $j > 1$ is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Proof. 1. **Assume $\mathbf{v}_j = c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}$.** Then $c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} - \mathbf{v}_j = \mathbf{0}$, with weight $-1 \neq 0$ on $\mathbf{v}_j$ (and weight 0 on the vectors after $\mathbf{v}_j$): a dependence relation.

2. **Assume the set is dependent**, with $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$, weights not all 0. Let j be the largest index with $c_j \neq 0$. If $j = 1$, then $c_1\mathbf{v}_1 = \mathbf{0}$ with $c_1 \neq 0$, so $\mathbf{v}_1 = \frac{1}{c_1}\mathbf{0} = \mathbf{0}$, which is false. So $j > 1$, and solving for $\mathbf{v}_j$ gives

$$\mathbf{v}_j = -\frac{1}{c_j}\left(c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}\right).$$

□

The facts from Week 3 hold in any vector space with the same proofs, since those proofs use only the vector space axioms: a set containing $\mathbf{0}$ is dependent ($1 \cdot \mathbf{0} = \mathbf{0}$ is a relation with a nonzero weight), and two vectors are dependent exactly when one is a multiple of the other.

Examples in function spaces.

- In $\mathbb{P}_n$, $\{1, t, \ldots, t^n\}$ is independent: two polynomials are equal exactly when their coefficients are equal, so $c_0 + c_1 t + \cdots + c_n t^n$ is the zero polynomial only if all $c_i = 0$. (Viewed as functions on $\mathbb{R}$, the same holds because a nonzero polynomial of degree at most n has at most n zeros; that fact from algebra is cited.)
- **Polynomials of different degrees are independent** (none being the zero polynomial). In a relation $c_1 p_1 + \cdots + c_k p_k = 0$ with some $c_i \neq 0$, let p_i have the largest degree d among those with $c_i \neq 0$. No other polynomial with a nonzero weight has degree d, so the coefficient of t^d in the sum is c_i times the leading coefficient of p_i, which is not 0. So the sum is not the zero polynomial, and every weight must be 0.
- $\{\sin t \cos t,\ \sin 2t\}$ is dependent in the space of functions on $\mathbb{R}$, since $\sin 2t = 2 \sin t \cos t$. $\{\sin t, \cos t\}$ is independent: if $\sin t = c \cos t$ for all t, then $t = 0$ gives $0 = c$, and then $\sin t = 0$ for all t, which is false; if $\cos t = c \sin t$ for all t, then $t = 0$ gives $1 = 0$.

Definition. A **basis** for a subspace H of V is a set $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ in H that is linearly independent and spans H.

- The columns $\mathbf{e}_1, \ldots, \mathbf{e}_n$ of I_n form the **standard basis** of $\mathbb{R}^n$. $\{1, t, \ldots, t^n\}$ is the standard basis of $\mathbb{P}_n$: it spans by definition of $\mathbb{P}_n$ and is independent by the example above.
- The columns of an invertible $n \times n$ matrix form a basis of $\mathbb{R}^n$: by the invertible matrix theorem they are independent and span $\mathbb{R}^n$.

Theorem (The spanning set theorem (Lay 4.3, Thm. 5))

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ be a set in V and $H = \operatorname{Span} S$.

(a) If some $\mathbf{v}_k$ in S is a combination of the other vectors of S, then removing $\mathbf{v}_k$ from S does not change the span.

(b) If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

Proof. 1. (a) Write $\mathbf{v}_k = \sum_{i \neq k} a_i \mathbf{v}_i$. In any combination $\sum_i c_i \mathbf{v}_i$, substitute this for $\mathbf{v}_k$: the result $\sum_{i \neq k} (c_i + c_k a_i) \mathbf{v}_i$ uses only the other vectors. So every vector of H lies in the span of S without $\mathbf{v}_k$, and that span is contained in H.

2. (b) If S is independent, it is a basis of H. If S is dependent and has two or more vectors, then by (W1) some vector of S is a combination of the others; remove it, which by (a) leaves the span equal to H.

3. Repeat step 2. Each removal leaves a smaller set that still spans H. If only one vector is left, it spans $H \neq \{\mathbf{0}\}$, so it is not $\mathbf{0}$, and one nonzero vector is independent. So the process stops at an independent subset of S that spans H: a basis.

□

A basis is as small and as large as possible.

- No vector can be removed from a basis $\mathcal{B}$ of H with the rest still spanning H: the removed vector would then be a combination of the others, and $\mathcal{B}$ would be dependent by (W1).
- No vector $\mathbf{u}$ of H can be added to $\mathcal{B}$ with the result still independent: $\mathbf{u}$ is a combination of the basis, so the enlarged set is dependent by (W1).

A basis for $\operatorname{Nul} A$: the vectors from the parametric vector form of $A\mathbf{x} = \mathbf{0}$, which span $\operatorname{Nul} A$ and are independent (W4).

Theorem (A basis for Col A (Lay 4.3, Thm. 6))

The pivot columns of a matrix A form a basis for Col A.

The idea

Row reduction changes the columns, so Col A and the column space of an echelon form are usually different. But it does not change which combinations of the columns are zero, since those combinations are the solutions of $A\mathbf{x} = \mathbf{0}$. In the reduced echelon form the dependence relations among the columns can be read off, and the same relations hold among the columns of A.

Proof. 1. Let B be the reduced echelon form of A. A list of weights $\mathbf{x}$ satisfies $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ exactly when $A\mathbf{x} = \mathbf{0}$, and by (W3) exactly when $B\mathbf{x} = \mathbf{0}$, that is, $x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n = \mathbf{0}$. So the columns of A and of B satisfy the same linear relations.

2. The pivot columns of B are distinct standard vectors $\mathbf{e}_i$, so they are independent.

3. Each non-pivot column $\mathbf{b}_j$ of B is a combination of the pivot columns to its left. Reason: an entry $b_{ij} \neq 0$ is not to the left of the leading 1 of row i (the leading entry is the first nonzero one), and it is not in that pivot's column, since $\mathbf{b}_j$ is not a pivot column; so $b_{ij} \neq 0$ only for rows i whose pivot lies to the left of column j. The pivot column of row i is $\mathbf{e}_i$, so $\mathbf{b}_j = \sum_i b_{ij}\mathbf{e}_i$ is a combination of those pivot columns.

4. By step 1, the pivot columns of A are independent and each non-pivot column of A is the same combination of pivot columns of A. By the spanning set theorem (a), removing the non-pivot columns does not change Col A.

□

Theorem (Row spaces (Lay 4.3))

If A and B are row equivalent, then Row A = Row B. If B is in echelon form, the nonzero rows of B form a basis for Row A.

Proof. 1. Each row operation replaces rows by combinations of rows, so the rows after the operation lie in the row space before it. Each operation can be undone by a row operation (W3), so the reverse inclusion holds too. So the row space never changes.

2. The nonzero rows of an echelon form B span Row B = Row A (zero rows add nothing).

3. They are independent. In a relation $c_1\mathbf{r}_1 + \cdots + c_k\mathbf{r}_k = \mathbf{0}$ among the nonzero rows, look at the column of the leading entry of $\mathbf{r}_1$: every later row has 0 there, so that entry of the sum is c_1 times the leading entry, and $c_1 = 0$. Then look at the leading column of $\mathbf{r}_2$ in the same way to get $c_2 = 0$, and so on down the rows.

□

The question answered. Row reduce A:

$$A \xrightarrow[R_4-5R_1]{R_2-3R_1,\ R_3-2R_1} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 1 & -1 & 4 \\ 0 & 0 & 2 & -2 & 13 \end{bmatrix} \xrightarrow[R_4-2R_2]{R_3-R_2} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix} \xrightarrow{R_4-\frac{3}{4}R_3} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and then $-\frac{1}{4}R_3$, $R_2 - 8R_3$, $R_1 + R_3$ give the reduced echelon form

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. The columns of B show $\mathbf{b}_2 = 4\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_1 - \mathbf{b}_3$, so columns 2 and 4 of A can be dropped: $\mathbf{a}_2 = 4\mathbf{a}_1$ and $\mathbf{a}_4 = 2\mathbf{a}_1 - \mathbf{a}_3$. The smallest spanning sets are bases.

2. Col A: the pivot columns of A (not of B), $\mathbf{a}_1 = (1, 3, 2, 5)$, $\mathbf{a}_3 = (0, 1, 1, 2)$, $\mathbf{a}_5 = (-1, 5, 2, 8)$.

3. Nul A: x_2, x_4 are free, $x_1 = -4x_2 - 2x_4$, $x_3 = x_4$, $x_5 = 0$. Basis $(-4, 1, 0, 0, 0)$, $(-2, 0, 1, 1, 0)$.

4. Row A: the nonzero rows of B, $(1,4,0,2,0)$, $(0,0,1,-1,0)$, $(0,0,0,0,1)$.

Shortcuts

- **Bases for the three spaces from one row reduction**: Col A: pivot columns of A; Row A: nonzero rows of an echelon form; Nul A: the vectors from the parametric form. Reason: the three theorems above. Instance: the question.
- **Basis of a subspace given as a span**: put the vectors in as columns and keep the pivot columns. Reason: Theorem 6. Instance: for $\{(1,2),(2,4),(0,1)\}$ the pivot columns of the matrix with these columns are 1 and 3, so $(1,2)$, $(0,1)$ is a basis of the span ($\mathbb{R}^2$).
- **A basis of** Row A **made of rows of** A: the pivot columns of A^T tell which rows of A to keep. Reason: Row A = Col A^T and Theorem 6. Instance: for the matrix A below, $R_2 - R_1$ reduces A^T as shown, so the pivot columns of A^T are 1 and 3, and rows 1 and 3 of A, $(1,1)$ and $(0,1)$, form a basis of Row A.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Polynomial subspaces**: write the conditions as equations on the coefficients, solve, and split by free coefficient. Reason: the coordinates of $a_0 + a_1 t + a_2 t^2$ are (a_0, a_1, a_2). Instance: $\{p \in \mathbb{P}_2 : p(0) = 0\}$ means $a_0 = 0$, so $p = a_1 t + a_2 t^2$, and $\{t, t^2\}$ is a basis (different degrees, so independent).

Traps

- The basis for Col A is the pivot columns *of* A. For the matrix below, the reduced echelon form has the column $(1,0)$, which is not even in Col A = Span$\{(1,2)\}$.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

- The basis for Row A is the nonzero rows *of the echelon form*, not the corresponding rows of A. For the matrix A of the shortcut "A basis of Row A made of rows of A", an echelon form has nonzero rows $(1,1)$, $(0,1)$ in rows 1 and 2, but rows 1 and 2 of A, $(1,1)$ and $(2,2)$, are dependent.

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{R_2 - 2R_1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- Row operations do not preserve dependence relations among the rows. The rows $\mathbf{r}_1 = (1,1)$ and $\mathbf{r}_2 = (2,2)$ satisfy $2\mathbf{r}_1 - \mathbf{r}_2 = \mathbf{0}$; after $R_2 - 2R_1$ the rows are $(1,1)$ and $(0,0)$, and $2(1,1) - (0,0) \neq \mathbf{0}$. (Relations among the *columns* are preserved, by step 1 of Theorem 6.)
- A basis is not unique: $\{(1,0),(0,1)\}$ and $\{(1,1),(1,-1)\}$ are both bases of $\mathbb{R}^2$.

§2.2 The dimension of a vector space (Lay 4.5)

The question. $\{(1,0),(0,1)\}$ and $\{(1,1),(1,-1)\}$ are both bases of $\mathbb{R}^2$. Must every basis of $\mathbb{R}^2$ have two vectors? Can a plane through $\mathbf{0}$ in $\mathbb{R}^3$ have a basis of three vectors? If all bases of a subspace have the same size, that number measures the subspace.

Theorem (More vectors than a basis are dependent (Lay 4.5, Thm. 9))
If a vector space V has a basis $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then any set in V containing more than n vectors is linearly dependent.

The idea

Write each of the vectors $\mathbf{u}_1, \dots, \mathbf{u}_p$ $(p > n)$ in terms of the basis. A dependence relation among the $\mathbf{u}$'s then comes from a homogeneous linear system with n equations (one per basis vector) and p unknowns (one weight per $\mathbf{u}_j$), and more unknowns than equations always give a nontrivial solution.

Proof. 1. Let $\mathbf{u}_1, \dots, \mathbf{u}_p$ be in V with $p > n$. Write $\mathbf{u}_j = a_{1j}\mathbf{b}_1 + \cdots + a_{nj}\mathbf{b}_n$, and let M be the $n \times p$ matrix with entries a_{ij}.

2. Since $p > n$, $M\mathbf{c} = \mathbf{0}$ has a nontrivial solution $\mathbf{c} = (c_1, \dots, c_p)$ (W2).

3. Then, regrouping by basis vector,

$$c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \sum_{j=1}^{p} c_j \sum_{i=1}^{n} a_{ij}\mathbf{b}_i = \sum_{i=1}^{n} \Big(\sum_{j=1}^{p} a_{ij}c_j\Big)\mathbf{b}_i = \sum_{i=1}^{n} (M\mathbf{c})_i\, \mathbf{b}_i = \mathbf{0},$$

with the c_j not all 0.

□

Theorem (All bases have the same size (Lay 4.5, Thm. 10))
If a vector space V has a basis of n vectors, then every basis of V has exactly n vectors.

Proof. 1. Let $\mathcal{B}$ be a basis of n vectors and $\mathcal{C}$ another basis, with m vectors.

2. $\mathcal{C}$ is independent, so by Theorem 9 applied to the basis $\mathcal{B}$, $m \le n$.

3. $\mathcal{B}$ is independent, so by Theorem 9 applied to the basis $\mathcal{C}$, $n \le m$. So $m = n$.

□

Definition. If V is spanned by a finite set, V is **finite-dimensional**, and the **dimension** $\dim V$ is the number of vectors in a basis of V; $\dim\{\mathbf{0}\} = 0$. Otherwise V is **infinite-dimensional**.

A finite-dimensional $V \neq \{\mathbf{0}\}$ has a basis by the spanning set theorem (b), and all its bases have the same size by Theorem 10, so $\dim V$ is well defined.

- $\dim \mathbb{R}^n = n$ (the standard basis) and $\dim \mathbb{P}_n = n + 1$ (the basis $1, t, \dots, t^n$).
- The space $\mathbb{P}$ of all polynomials is infinite-dimensional: a finite set of polynomials has a largest degree d, its span contains only polynomials of degree at most d, and t^{d+1} is not among them.
- **Bounds.** If p vectors span V, then $\dim V \le p$, because they contain a basis (spanning set theorem). If V contains p independent vectors, then $\dim V \ge p$, because otherwise Theorem 9 would make them dependent.

Worked example: dimension of a subspace given by parameters. Let $H = \{(a+2b,\ b-c,\ a+2c,\ 3a+6b)\}$ in $\mathbb{R}^4$, with a, b, c real.

1. Split by parameter: the vector is $a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$ with $\mathbf{v}_1 = (1, 0, 1, 3)$, $\mathbf{v}_2 = (2, 1, 0, 6)$, $\mathbf{v}_3 = (0, -1, 2, 0)$, so $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
2. $\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2$, so by the spanning set theorem $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Neither of these is a multiple of the other (look at entries 2 and 3), so they are independent.
3. So $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis and $\dim H = 2$, even though H was written with three parameters.

Theorem (Subspaces of a finite-dimensional space (Lay 4.5, Thm. 11))

Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be extended, if necessary, to a basis for H. Also, H is finite-dimensional and $\dim H \leq \dim V$.

Proof. 1. If $H = \{\mathbf{0}\}$, then $\dim H = 0 \leq \dim V$. Otherwise let S be an independent set in H (if none is given, start with one nonzero vector of H).

2. **Assume S does not span H.** Choose $\mathbf{u}$ in H outside $\operatorname{Span} S$. Then $S \cup \{\mathbf{u}\}$ is independent: in a relation, the weight on $\mathbf{u}$ must be 0 (otherwise $\mathbf{u}$ would be a combination of S), and then all the other weights are 0 because S is independent.

3. Repeat step 2. Every set produced is independent in V, so by Theorem 9 it has at most $\dim V$ vectors. So the process stops, and it stops only when the set spans H: a basis of H with at most $\dim V$ vectors.

□

Theorem (The basis theorem (Lay 4.5, Thm. 12))

Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p vectors in V is a basis for V. Any set of exactly p vectors that spans V is a basis for V.

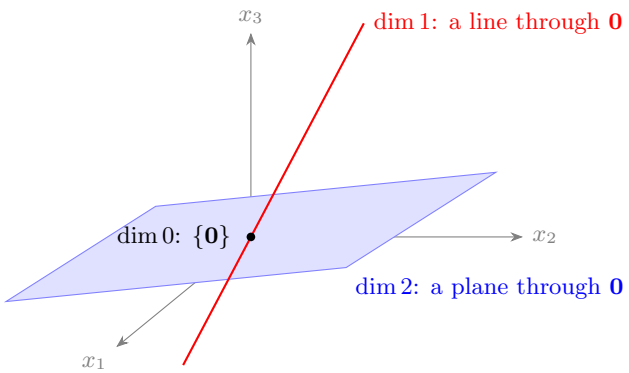
Proof. 1. **Assume S is independent with p vectors.** By Theorem 11 (with $H = V$), S extends to a basis of V, which has exactly p vectors (Theorem 10). So nothing was added, and S is a basis.

2. **Assume S spans V with p vectors.** By the spanning set theorem, S contains a basis of V, which has exactly p vectors. So nothing was removed, and S is a basis.

□

Subspaces of $\mathbb{R}^3$ by dimension. A subspace H of $\mathbb{R}^3$ has $\dim H \leq 3$ by Theorem 11.

- $\dim H = 0$: $H = \{\mathbf{0}\}$.
- $\dim H = 1$: $H = \operatorname{Span}\{\mathbf{v}\}$ with $\mathbf{v} \neq \mathbf{0}$, a line through $\mathbf{0}$.
- $\dim H = 2$: $H = \operatorname{Span}\{\mathbf{u}, \mathbf{v}\}$ with $\mathbf{u}, \mathbf{v}$ independent, a plane through $\mathbf{0}$.
- $\dim H = 3$: a basis of H is 3 independent vectors in $\mathbb{R}^3$, which by the basis theorem is a basis of $\mathbb{R}^3$; so $H = \mathbb{R}^3$.



The proper subspaces of $\mathbb{R}^3$: the origin, lines through the origin, and planes through the origin. The only 3-dimensional subspace is $\mathbb{R}^3$ itself.

The question answered. Every basis of $\mathbb{R}^2$ has 2 vectors (Theorem 10). A plane through $\mathbf{0}$ in $\mathbb{R}^3$ is $\operatorname{Span}\{\mathbf{u}, \mathbf{v}\}$ with $\mathbf{u}, \mathbf{v}$ independent, so it has dimension 2; by Theorem 9 any three vectors in it are dependent, so it has no basis of 3 vectors.

Shortcuts

- **Is this set a basis of $\mathbb{R}^n$?** With exactly n vectors, check independence only (or $\det \neq 0$). With fewer than n: no, since by the bounds above they cannot span. With more than n: no, since they are dependent

(Theorem 9). Reason: the basis theorem. Instance: $(1,2)$, $(3,4)$ have det $= 4-6 \neq 0$, so they are a basis of $\mathbb{R}^2$.

- **Extending to a basis of $\mathbb{R}^n$**: row reduce $[\,\mathbf{v}_1 \cdots \mathbf{v}_k\ \mathbf{e}_1 \cdots \mathbf{e}_n\,]$ and keep the pivot columns. Reason: the columns span $\mathbb{R}^n$ (they include the $\mathbf{e}_i$), and by step 3 of Theorem 6 a column fails to be a pivot column only if it is a combination of earlier columns, which never happens for the independent $\mathbf{v}$'s. Instance: for $(1,1,0)$, row reduction gives pivot columns $1,2,4$, so the basis is $(1,1,0)$, $\mathbf{e}_1$, $\mathbf{e}_3$.

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Dimension of a span**: the number of pivot columns of the matrix with those columns. Reason: Theorem 6. Instance: the worked example has $\dim H = 2$.

Traps

- The basis theorem needs the right count: two independent vectors in $\mathbb{R}^3$, such as $\mathbf{e}_1, \mathbf{e}_2$, are not a basis of $\mathbb{R}^3$.
- $\dim \mathbb{R}^2 = 2$, but $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$; a plane through $\mathbf{0}$ in $\mathbb{R}^3$ is a 2-dimensional subspace of $\mathbb{R}^3$.
- A dependent set gives no bound on the dimension: $\{\mathbf{0}, \mathbf{e}_1\}$ is a dependent set of 2 vectors in $\mathbb{R}^3$, which has dimension 3.
- The number of parameters is not the dimension: the worked example uses three parameters and has dimension 2.

§2.3 Rank (Lay 4.5)

The question. A 7×9 matrix has a two-dimensional null space. Without seeing the matrix, what is the dimension of its column space, and is Col A all of $\mathbb{R}^7$?

- $\dim \operatorname{Nul} A$ is the number of free variables, because the parametric vector form gives a basis of Nul A with one vector per free variable (W4).
- $\dim \operatorname{Col} A$ is the number of pivot columns (Theorem 6).
- $\dim \operatorname{Row} A$ is the number of pivots, the number of nonzero rows of an echelon form (row space theorem).

Definition. The **rank** of A is $\operatorname{rank} A = \dim \operatorname{Col} A$. The **nullity** is $\dim \operatorname{Nul} A$.

By the three items above, $\operatorname{rank} A = \dim \operatorname{Row} A$, and since $\operatorname{Row} A = \operatorname{Col} A^T$, also $\operatorname{rank} A = \operatorname{rank} A^T$.

Theorem (The rank theorem (Lay 4.5))
For an $m \times n$ matrix A, $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

Proof. 1. Each of the n columns of A is either a pivot column or the column of a free variable, and not both.

2. The pivot columns number rank A (Theorem 6) and the free variables number $\dim \operatorname{Nul} A$ (first item above). So the two numbers add up to n.

□

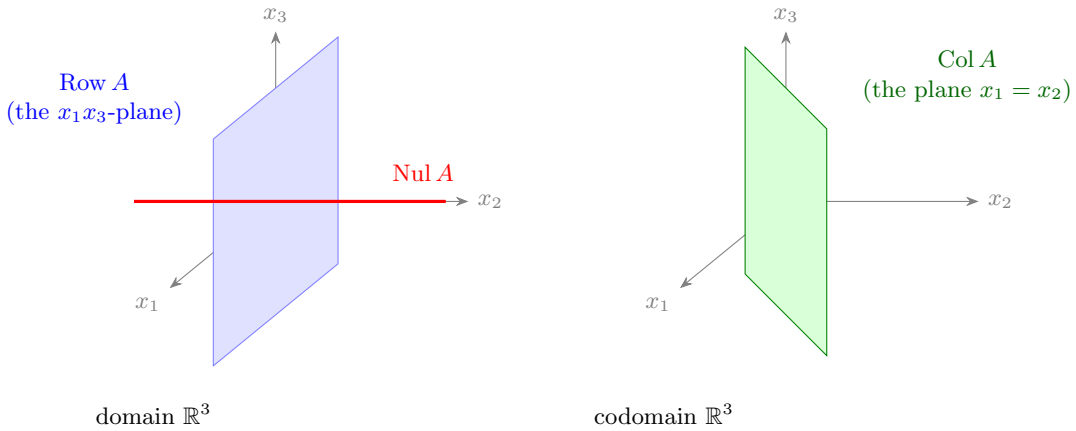
The question answered. By the rank theorem, $\operatorname{rank} A = 9 - 2 = 7$. So Col A is a 7-dimensional subspace of $\mathbb{R}^7$; a basis of it is 7 independent vectors in $\mathbb{R}^7$, hence a basis of $\mathbb{R}^7$ (basis theorem), and Col $A = \mathbb{R}^7$.

Worked example: consistency from dimensions. A homogeneous system of 50 equations in 54 unknowns has a solution set spanned by 4 independent vectors. Is every system with the same coefficients and any right-hand side consistent? The coefficient matrix A is 50×54 with $\dim \operatorname{Nul} A = 4$, so $\operatorname{rank} A = 54 - 4 = 50$, and as above $\operatorname{Col} A = \mathbb{R}^{50}$. So $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ (W2).

The invertible matrix theorem, continued (Lay 4.5). For an $n \times n$ matrix A, each of the following is equivalent to A being invertible: (m) the columns of A form a basis of $\mathbb{R}^n$; (n) $\operatorname{Col} A = \mathbb{R}^n$; (o) $\dim \operatorname{Col} A = n$; (p) $\operatorname{rank} A = n$; (q) $\operatorname{Nul} A = \{\mathbf{0}\}$; (r) $\dim \operatorname{Nul} A = 0$. Reason: (m) is "independent and spanning"; (n), (o), (p) each say there are n pivot columns, using the basis theorem for (o)⇒(n); (q) is "$A\mathbf{x} = \mathbf{0}$ only trivially"; (r) is (q) restated; (p) $\iff$ (r) is the rank theorem.

The three spaces in a picture. The figure shows $\operatorname{Row} A$, $\operatorname{Nul} A$ and $\operatorname{Col} A$ for

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 3 & 0 & -1 \\ 4 & 0 & 5 \end{bmatrix}.$$



The three spaces of the matrix A above. Rank 2: $\operatorname{Row} A$ and $\operatorname{Col} A$ are planes, and $\operatorname{Nul} A$ is a line ($2 + 1 = 3$ columns). $\operatorname{Nul} A$ is the x_2-axis, because the equations $3x_1 - x_3 = 0$ and $4x_1 + 5x_3 = 0$ force $x_1 = x_3 = 0$.

Shortcuts

- **Counting**: $\operatorname{rank} A$ is the number of pivots, which equals $\dim \operatorname{Col} A$ and $\dim \operatorname{Row} A$; and $\dim \operatorname{Nul} A = n - \operatorname{rank} A$. Reason: the rank theorem. Instance: the question.
- **Rank bounds**: $\operatorname{rank} A \leq \min(m, n)$, since there is at most one pivot per row and per column (W5). Instance: a 6×9 matrix has rank at most 6, so its null space has dimension at least $9 - 6 = 3$.
- **Full rank m means onto**: if $\operatorname{rank} A = m$ (the number of rows), then $\operatorname{Col} A = \mathbb{R}^m$ and $A\mathbf{x} = \mathbf{b}$ is always consistent. Reason: basis theorem. Instance: the 50×54 example.

Traps

- The rank theorem has n (the number of columns), not m: for the question's 7×9 matrix, $7 + 2 = 9$.
- $\dim \operatorname{Row} A = \dim \operatorname{Col} A$, but the spaces themselves differ in general: for the matrix below, $\operatorname{Row} A = \operatorname{Span}\{(0, 1)\}$ and $\operatorname{Col} A = \operatorname{Span}\{(1, 0)\}$.

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- $\operatorname{Nul} A$ and $\operatorname{Col} A$ live in different spaces when $m \neq n$: for a 3×7 matrix of rank 3, $\operatorname{Nul} A$ is a 4-dimensional subspace of $\mathbb{R}^7$, not $\mathbb{R}^4$.

§3 Homework this week

HW5, Section 4.3, and HW6, Section 4.5.

Section 4.3.

- **#3, #5** (bases of $\mathbb{R}^3$?). Row reduce the matrix with these columns: a pivot in every column gives independence, a pivot in every row gives spanning; a set containing $\mathbf{0}$ is dependent.
- **#13** (bases for $\operatorname{Col} A$, $\operatorname{Row} A$, $\operatorname{Nul} A$ from $A \sim B$). Pivot columns of A; nonzero rows of B; the parametric form (as in the question).
- **#15** (basis of a span). Vectors as columns; keep the pivot columns.
- **#20** (a given relation). Drop a vector whose weight in the relation is nonzero (spanning set theorem); the remaining two are not multiples of each other.
- **#23, #24, #26, #29, #31** (true/false). A basis needs independence and spanning; the spanning set theorem (b); a basis is a largest independent set; $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solutions; the trap on row relations.
- **#41** (dependent images). Apply T to a nontrivial relation and use linearity.
- **#44** (polynomials p_1, p_2, p_3). Find the relation among them, then the spanning set theorem.

Section 4.5.

- **#1, #3** (dimension of a subspace given by parameters). Split by parameter, remove redundant vectors (worked example).
- **#9, #11** (dimension of a span; of $\operatorname{Nul} A$, $\operatorname{Col} A$, $\operatorname{Row} A$). Count pivot columns and free variables.
- **#17, #18, #22, #24, #25** (true/false). Theorem 6; nullity is the number of free variables; both ranks are the number of pivots; the rank theorem; the bound from a spanning set.
- **#27** (Hermite polynomials). Different degrees, so independent; then the basis theorem in $\mathbb{P}_3$.
- **#33, #35** (counting). The rank theorem and $\operatorname{rank} A = \operatorname{rank} A^T$; a 4-dimensional subspace of $\mathbb{R}^4$ is $\mathbb{R}^4$, and $\operatorname{Nul} A$ lives in $\mathbb{R}^n$.
- **#44, #45** (true/false on bounds). A dependent set gives no bound (third trap of 4.5); extend an independent set (Theorem 11).

§4 Problem set

P1. For the matrix below, find bases for $\operatorname{Col} A$, $\operatorname{Row} A$ and $\operatorname{Nul} A$, and check the rank theorem.

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 3 & 6 & 0 & 3 \end{bmatrix}$$

P2. Is $\{(1, 0, -1), (2, 1, 0), (1, 3, 4)\}$ a basis of $\mathbb{R}^3$?

P3. Find a basis for, and the dimension of, $H = \operatorname{Span}\{(1, 0, 2), (2, 1, 5), (0, -1, -1)\}$.

P4. Find a basis for $\{p \in \mathbb{P}_3 : p(1) = 0\}$ and its dimension.

P5. (a) A is 5×8 with rank 5. What is $\dim \operatorname{Nul} A$? Is $\operatorname{Col} A = \mathbb{R}^5$? (b) Can a 4×6 matrix have a one-dimensional null space?

P6. Extend $\{(1, 2, 0), (0, 1, 3)\}$ to a basis of $\mathbb{R}^3$.

P7. True or false, with a reason. (a) If $\dim V = 3$, any 4 vectors in V are dependent. (b) If $\dim V = 3$, any 2 vectors in V span a 2-dimensional subspace. (c) The number of pivot columns of A equals $\dim \operatorname{Row} A$. (d) If A is 3×3 and $\dim \operatorname{Nul} A = 1$, then $\det A = 0$.